

ANNEXURE A – TRANSFORMER MECHANICS WORKSHEET

SET – C

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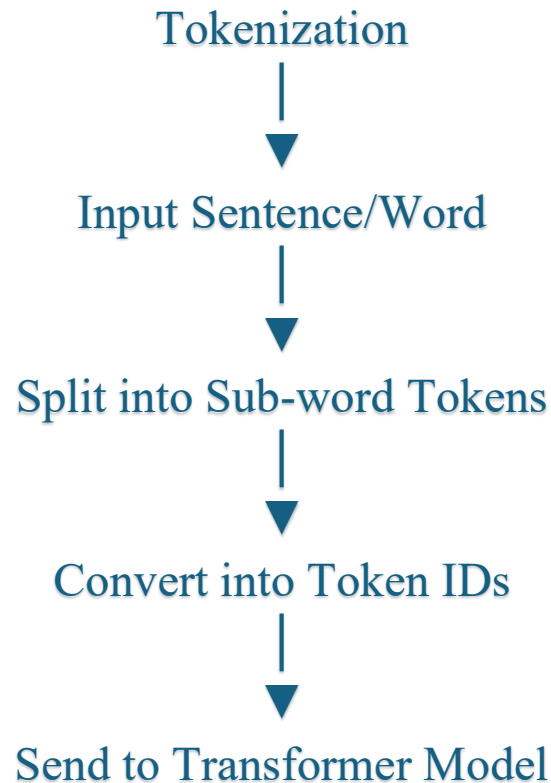
Course Name:Generative-AI

Question 1: Tokenization

Theory:

Tokenization is the process of breaking text into smaller units called tokens. Transformer models usually use sub-word tokenization (such as BPE or WordPiece), where long or uncommon words are divided into meaningful parts instead of treating the entire word as one token. This helps the model understand rare or unseen words

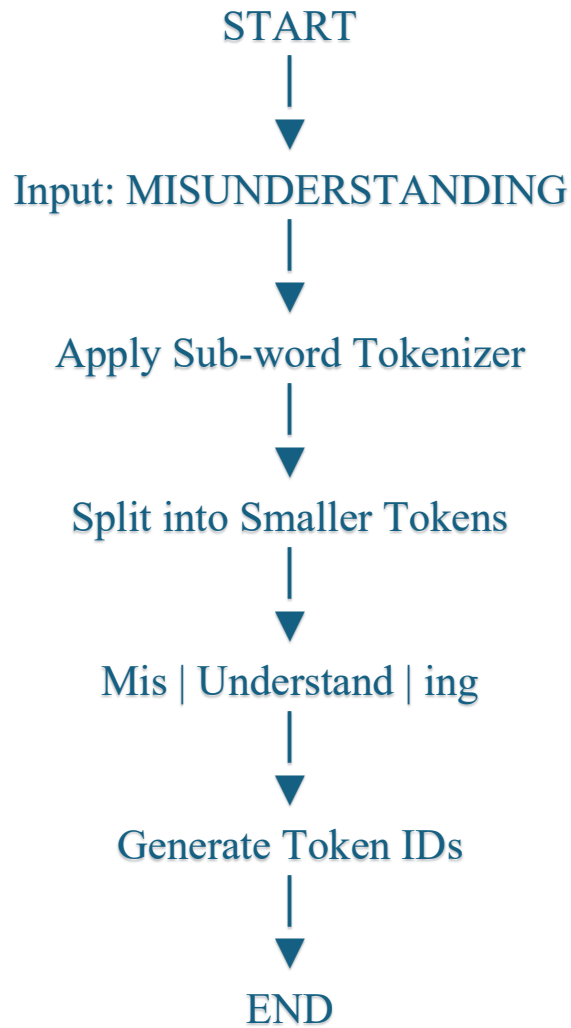
Structure



Explanation:

- **Input Word/Sentence:** Original text given to the model.
- **Split into Tokens:** Breaks text into smaller parts.
- **Sub-word Token Generation:** Creates meaningful sub-words.
- **Convert to Token IDs:** Converts tokens into numerical IDs.
- **Input to Transformer:** IDs are sent to the Transformer for processing.

Flowchart



(a)MISUNDERSTANDING



Mis | Understand | ing

Justification

- **Mis** → Prefix
- **Understand** → Root word

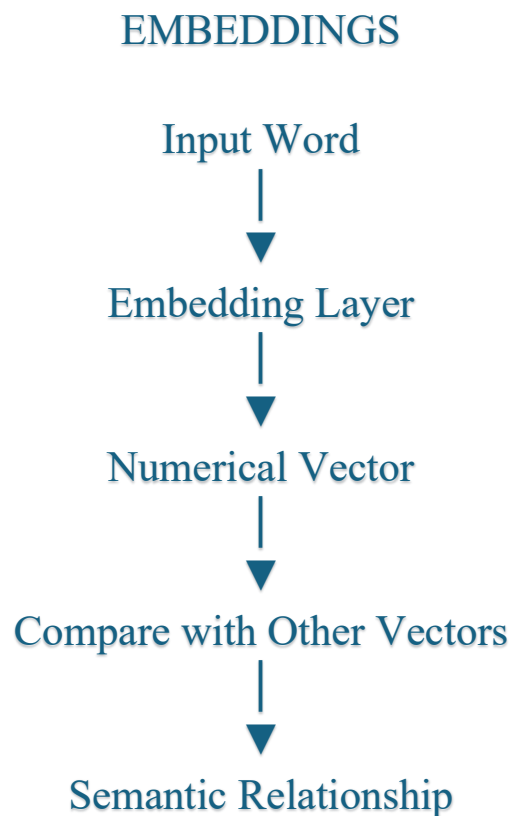
- **Ing** → Suffix

(b) Sub-word tokenization helps the model understand rare or unseen words by breaking them into smaller known parts. This reduces vocabulary size and improves language understanding.

Question 2: Embeddings

Theory: A word embedding is a dense numerical vector that represents the meaning of a word. Similar words have similar vectors. Cosine similarity measures how close two word vectors are.

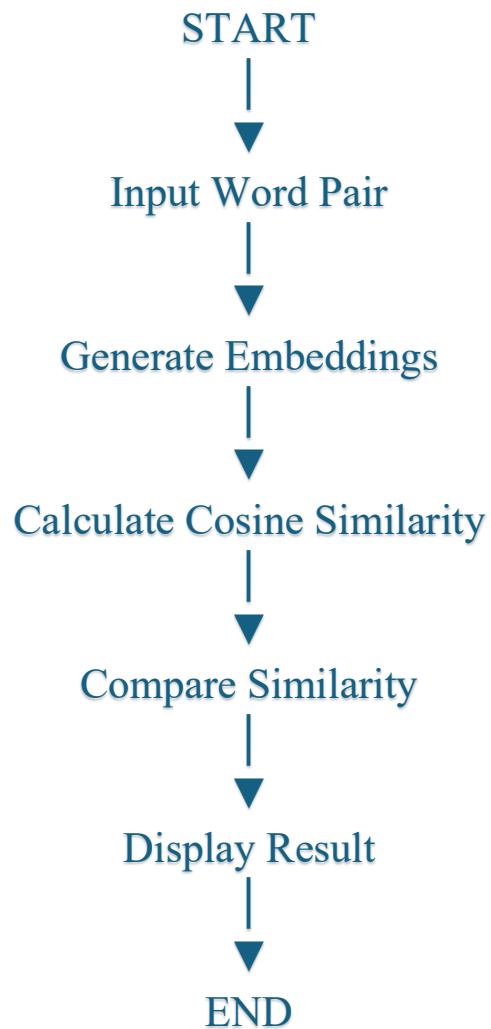
Structure



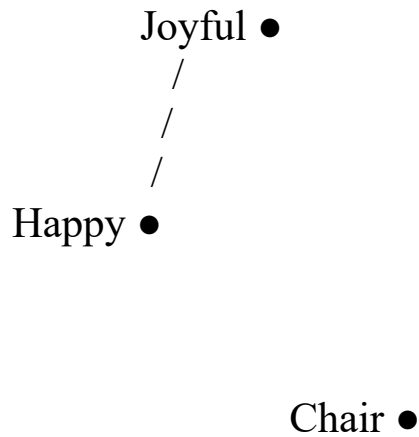
Explanation

- **Input Word:** Word given to the model.
- **Embedding Layer:** Converts the word into numbers.
- **Numerical Vector:** Represents meaning.
- **Compare Vectors:** Measures similarity.
- **Semantic Relationship:** Finds words with similar meanings.

Flowchart



Embedding Space



(a) **Happy** and **Joyful** have higher cosine similarity because both words have similar meanings.

(b) An embedding is a dense numerical representation of a word that captures its meaning.

A one-hot vector only identifies the position of the word and does not represent semantic meaning.

Embedding	One-Hot Vector
Dense	Sparse
Represents meaning	Represents identity
Similar words are close	No similarity

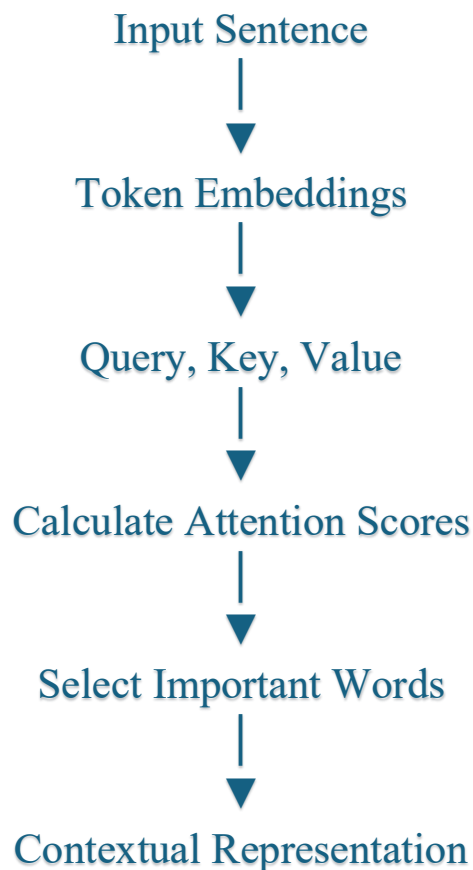
Question 3: Self-Attention

Sentence:

"Sam tried to call George on the phone, but he wasn't successful."

Theory: Self-attention is the mechanism that allows each word to look at every other word in the sentence. It helps the Transformer understand context and identify relationships between words.

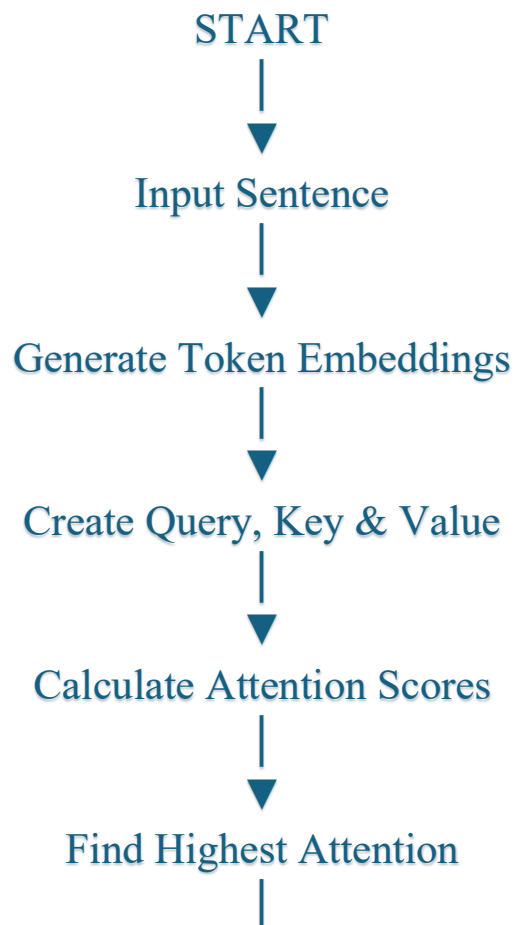
Structure

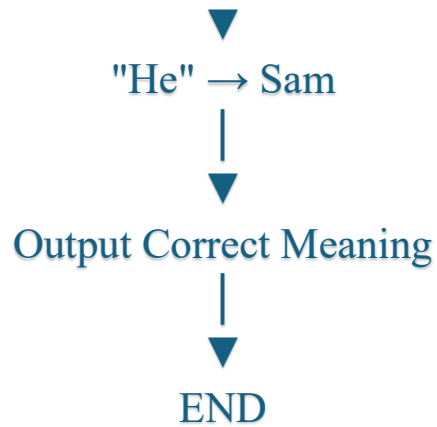


Explanation

- **Input Sentence:** Sentence entered into the Transformer.
- **Token Embeddings:** Converts words into vectors.
- **Query, Key, Value:** Used to compare relationships between words.
- **Attention Scores:** Calculates how much each word should focus on another.
- **Select Important Words:** Chooses the most relevant words.
- **Contextual Representation:** Produces the correct meaning.

Flowchart





Attention Diagram

Self-Attention

Sam -----▶ he ✓

George -----▶ he ✗

Highest Attention = Sam

(a) ✓ Sam

(b) Self-attention examines all words in the sentence before deciding the meaning of a word. It understands that Sam is the person trying to make the phone call, so "he" refers to Sam. A simple word-order rule may incorrectly choose George, who appears closer to the pronoun.

Question 4: Query, Key, Value

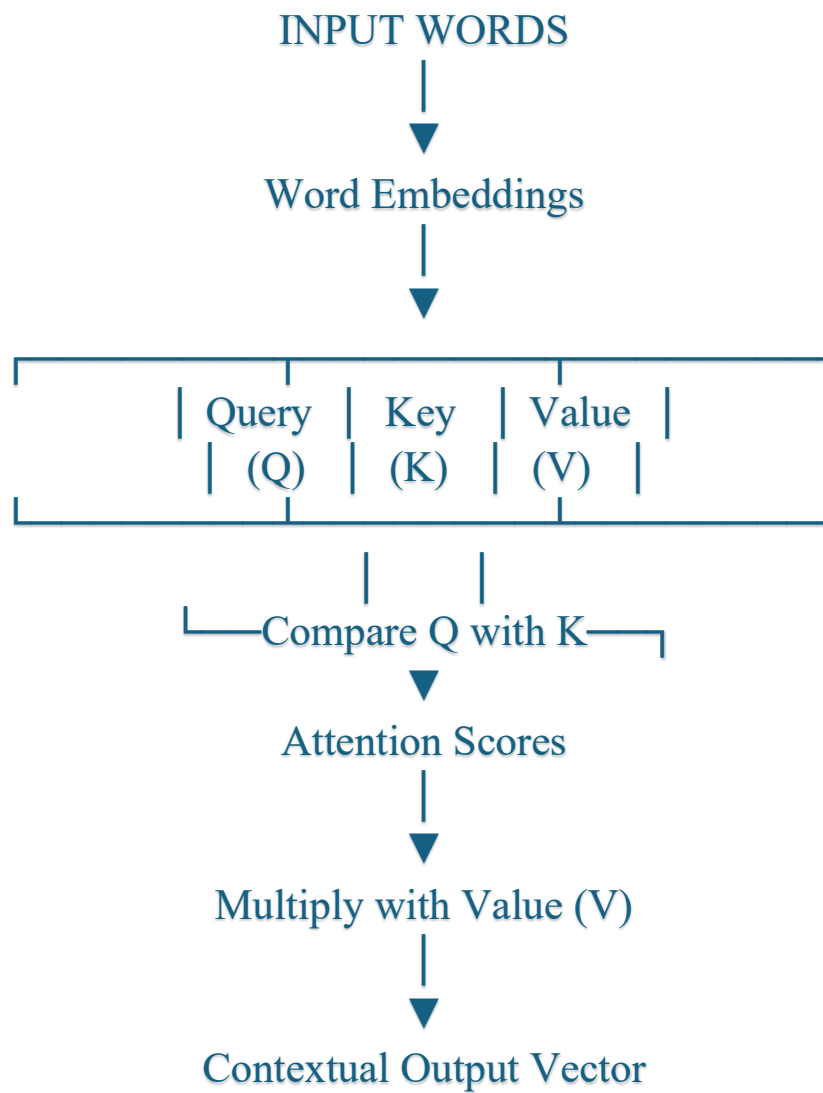
Theory:

In the Transformer self-attention mechanism, every input word is converted into three vectors:

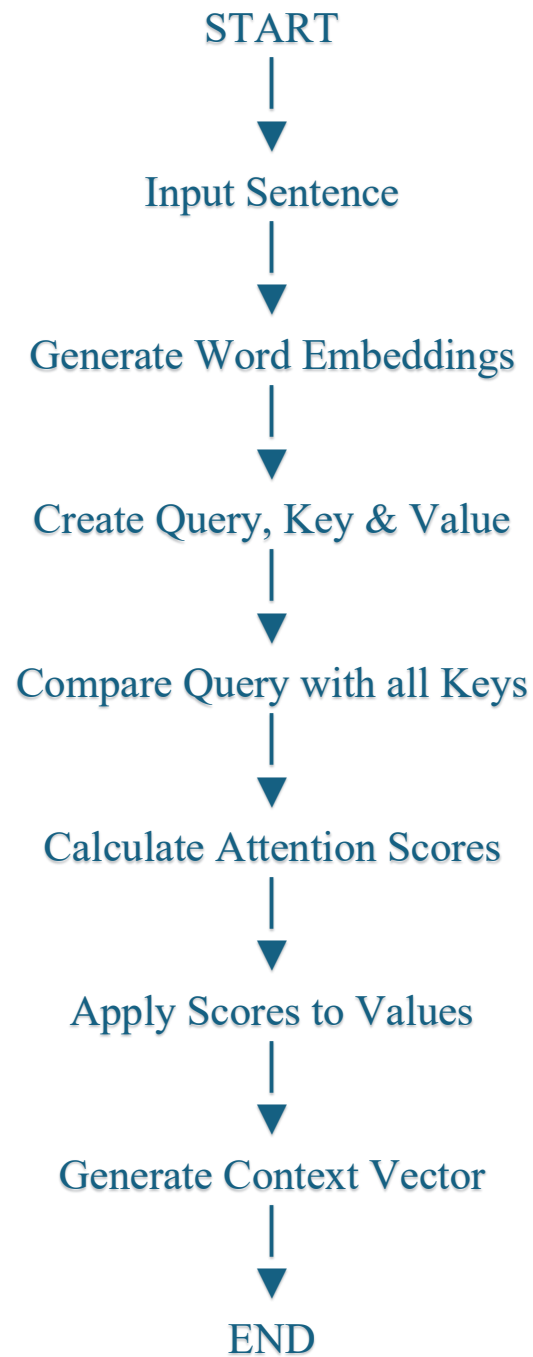
- **Query (Q)**
- **Key (K)**
- **Value (V)**

These vectors help the model determine which words are important and how much attention should be given to each word while processing a sentence.

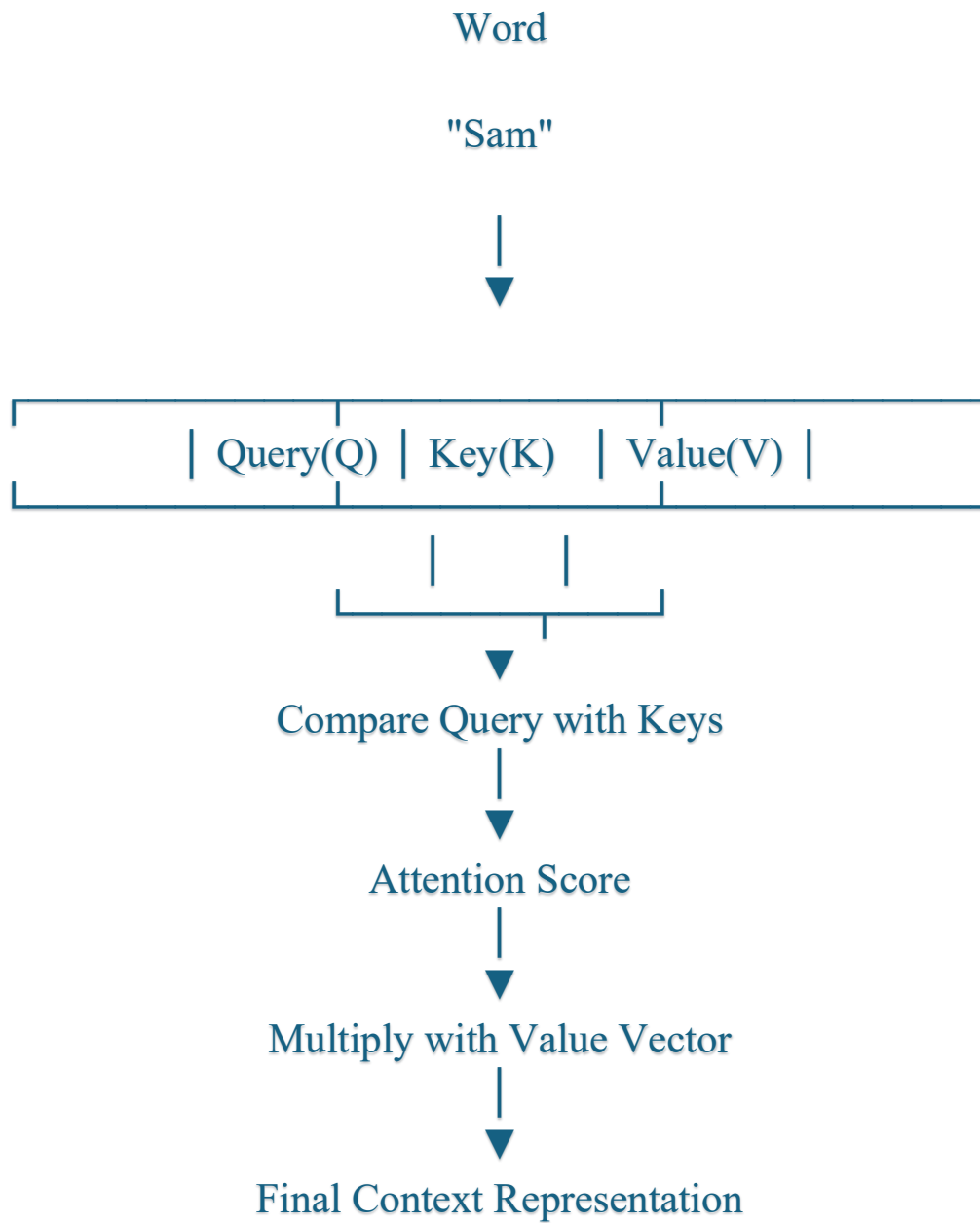
Structure (Query–Key–Value Mechanism)



Flowchart



Working Diagram



Query (Q)

The Query vector represents what the current word is looking for. It is used to search for relevant information from other words in the sentence.

Key (K)

The Key vector represents the information each word contains. It is compared with the Query vector to determine how relevant a word is.

Value (V)

The Value vector contains the actual information or meaning of the word. Once attention scores are calculated, the Value vectors are combined to produce the final output.

How do they produce an attention score?

1. Each word is converted into Query, Key, and Value vectors.
2. The Query of the current word is compared with the Keys of all words.
3. Similarity between Query and Key gives an attention score.
4. The scores are normalized using Softmax.
5. These scores are multiplied with the Value vectors.
6. The weighted Values are added together to produce the contextual output.

Simple Example

"Sam likes ice cream."

Word: Sam

Query → Looking for related words



Compare with Keys of:

Sam
Likes
Ice
Cream



Attention Scores

Sam = 0.10

Likes = 0.60

Ice = 0.20

Cream = 0.10



Multiply by Value Vectors



Final Context Representation

Summary Table

Vector	Purpose
Query (Q)	Finds relevant words
Key (K)	Matches with Query to calculate attention
Value (V)	Provides the information used to generate the final output

Conclusion

The Query, Key, and Value (QKV) mechanism is the core of the self-attention process in Transformers. Query searches for relevant information, Key determines how relevant each word is, and Value provides the actual information. Together, they compute attention scores that help the Transformer focus on the most important words in a sentence, enabling accurate context understanding and better language processing.